Online Supplement

Source-Scored Structural Initial Designs for Chance-Constrained Optimization of Ordered Policy Profiles

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A Finite Algorithm

Algorithm 1: Source-scored structural initial design

1. Generate the declared 64-profile outcome-free library on 128 regular midpoints using seed 20260808.
2. For every source task and profile, compute the replicated objective mean and the Gaussian chance-margin statistic in (8).
3. Convert each statistic to deterministic within-task average-tie percentile ranks; average the ranks uniformly across source tasks.
4. Compute exact Voronoi-cell cosine coefficients $c_0, \ldots, c_8$, divide c_k by $1 + .25k$, append all nine diagonal squares, and standardize the 18 coordinates over the library.
5. Append source safety and objective ranks. Select the first center by (11); select nine further centers by (12).
6. Linearly interpolate each selected profile to the declared target grid, clip to bounds, and deterministically round. Freeze these ten policies before the first target outcome.
7. Run any declared target backend. Freeze a distinct ordered shortlist before verification.
8. Test each shortlist candidate on its independent all-success stream. Deploy the first certificate; abstain if none certifies.

The stable tie order is safety rank, objective rank, profile identifier for the first center and finite-library index for farthest-first. Robust source feasibility is diagnostic only. There is no endpoint replacement, hidden scalarization, target-outcome repair, or target-oracle branch in a non-oracle arm.

B Additional Statistical Results

B.1 Verifier thresholds

Table S1 gives exact power calculations. At $n = 80$, the Clopper–Pearson lower-bound rule equals the all-success rule. At $n = 160$, its preregistered sensitivity threshold permits two failures. The primary results retain the all-success protocol.

Table S1: Exact candidate-wise certificate power at budgets 80 and 160.

n	p	All-success power	CP threshold	Allowed failures	CP power
80	0.950	0.017	80	0	0.017
80	0.975	0.132	80	0	0.132
80	0.990	0.448	80	0	0.448
80	0.995	0.670	80	0	0.670
80	1.000	1.000	80	0	1.000
160	0.950	0.000	158	2	0.012
160	0.975	0.017	158	2	0.234
160	0.990	0.200	158	2	0.784
160	0.995	0.448	158	2	0.953
160	1.000	1.000	158	2	1.000

B.2 Full sensitivity matrix

The full one-factor sensitivity matrix is reported in Table S2. Entries are ordered by the tested levels in the second column. Effective-rank levels are explicit overrides; all other rows include the frozen baseline level. The nonmonotone responses, the zero certificate rate at $\alpha = .01$, and the immaterial frequency penalty are retained rather than optimized away.

Table S2: One-factor sensitivity of the source-scored design at $d = 1000$. Each rate aggregates eight fixed regime strata and 20 independently seeded latent tasks per regime at one resolution.

Factor	Tested levels	Feasible (%)	Certified (%)
Effective rank	4/8/16/32	89.4/85.0/84.4/83.8	55.6/36.3/25.6/29.4
Chance level α	.01/.05/.1	95.0/95.0/95.0	0.0/46.2/29.4
Calibrated safe mass	.03/.08/.18	69.4/95.0/99.4	22.5/46.2/70.0
Initial-design size n_0	5/10/20	79.4/95.0/96.9	42.5/46.2/46.9
Source tasks	1/2/4	85.0/95.0/95.6	33.8/46.2/45.0
Profiles per source task	32/64/128	83.1/95.0/73.8	38.1/46.2/38.8
Source replications	1/3/5	92.5/95.0/92.5	45.0/46.2/45.6
Retained frequency K	4/8/16	81.9/95.0/95.6	43.1/46.2/48.1
Frequency penalty κ	0/.25/1	95.0/95.0/95.0	46.2/46.2/46.2
Safety-rank weight	.25/1/4	95.0/95.0/95.0	46.9/46.2/46.2
Objective-rank weight	.25/1/4	95.0/95.0/95.0	46.9/46.2/46.2
First-center safety weight	.25/.5/.75	62.5/95.0/95.0	32.5/46.2/45.0

Recorded wall time is descriptive rather than a resource-normalized endpoint. On the mixed CPU cluster, the primary ten-point initial-design arms average 1.2–1.7 seconds per synthetic task, whereas 394-call source-free controls average about 59 seconds and the 394-call functional-SCBO arm has a median of about 677 seconds. The synthetic simulator is analytic and cheap, so these figures measure implementation overhead rather than the time of a physical simulator; simulation-call accounting remains the portable comparison.

B.3 Task-seed strata audit

The registered generator includes a deterministic `task_seed mod 5` cosine component. Table S3 reports its realized categories rather than silently treating the finite sample as a balanced IID mixture. The analysis uses fixed regime strata and applies concentration separately at each resolution.

Table S3: Realized latent-category counts among the 160 independent task seeds. The same seeds are crossed with all three grid resolutions.

Regime	Category 0	1	2	3	4	Total
Aligned low frequency	3	8	3	1	5	20
Growing rank	4	7	6	3	0	20
Frequency shift	8	2	3	2	5	20
Permutation	11	0	3	5	1	20
Irregular grid	4	4	6	4	2	20
Piecewise smooth	5	4	6	4	1	20
Sparse high frequency	3	1	7	4	5	20
Misspecified target	3	6	6	2	3	20
Overall	41	32	40	25	22	160

B.4 Native end-to-end transfer

Table S4: Native transfer pipelines at $N = 20$ with the same 384-call source archive and independent verifier. Each method retains its own initialization and online adaptation.

Native method	FactorShock	Inventory	Queue	Total	False
Safe F-PACOH	0/20	0/20	0/20	0/60	0
RGPE	14/20	16/20	0/20	30/60	0
Stacked GP	0/20	13/20	0/20	13/60	0
MTGP	1/20	0/20	0/20	1/60	0
FSBO	6/20	2/20	0/20	8/60	0
HyperBO	0/20	0/20	0/20	0/60	0
MetaBO	19/20	9/20	0/20	28/60	0
MALIBO	12/20	2/20	0/20	14/60	0

No method certifies a Queue task. These 60-task totals measure repeatability on three fixed families, not generalization to a population of domains.

B.5 Energy temporal audit

Table S5: Postdecision Energy temporal stability. Counts are descriptive and do not replace the declared empirical-window certificate.

Matrix	Originally certified	Block stable	Nonoverlap stable	Joint stable
Energy V2	317	303/317	316/317	303/317
Functional V2	62	59/62	61/62	58/62
Energy V3	358	312/358	358/358	312/358

V3 preserves all 358 original certificates under physically nonoverlapping windows but 312 under the joint block criterion. Shared calendars and only five regions preclude an iid market-population interpretation.

C Secondary Cumulative-Variance Diagnostic

The project originally studied cumulative heteroscedastic variance decomposition. Under matched policies, the factor-HVD model represents local and shared exposure and improves held-out variance-shape estimation. Table S6 gives the final matched diagnostic.

Table S6: Pooled variance versus cumulative factor-HVD. HVD improves calibration but not feasible recovery or verification cost.

Domain	Variance model	Feasible	Log-RMSE	Shape corr.	Mean verify
FactorShock	Pooled	20/20	1.158	0.000	159.2
FactorShock	Cumulative factor-HVD	20/20	0.219	0.919	158.4
Inventory	Pooled	20/20	0.577	0.000	182.4
Inventory	Cumulative factor-HVD	20/20	0.349	0.953	184.0
Queue	Pooled	19/20	0.501	0.000	190.4
Queue	Cumulative factor-HVD	19/20	0.254	0.995	197.6

Because both variance models recover the same feasible counts and HVD slightly increases verification cost in Inventory and Queue, HVD is not included in the paper's claimed optimization method. Its formal results remain in the proof archive as reusable mechanistic results.

D Machine-Checked Proof Map

The Lean 4 project builds against mathlib without `sorry`, `admit`, or project-defined `axiom`. Table S7 maps manuscript claims to their principal files. Lean proves each displayed implication; source-target alignment and simulator-distribution assumptions remain scientific premises.

Table S7: Principal Lean files under `proof/SCOLHKG/`.

Claim	File
Finite-atlas no-free-lunch	`Real/ProposalNoFreeLunch.lean`
Profile-coordinate and interpolation error	`Real/ProfileCoordinateConsistency.lean`
Farthest-first factor-two coverage	`Real/FarthestFirstKCenter.lean`
Finite mean/scale margin and rank recovery	`Measure/SourceRankRecovery.lean`, `Real/SourceRankRecovery.lean`
Stratified, non-identical task calibration	`Measure/TaskAtlasCoverage.lean`
Augmented-to-structural projected coverage	`Real/GeometricAtlasCoverage.lean`
Exact all-success candidate validity	`Measure/ExactBinomialCertificate.lean`
Composed frontend and budget interfaces	`Real/PaperMainline.lean`

E Reproducibility Contract

The final evidence registry contains 14 matrix contracts, 12 compact analyses, and SHA-256 commitments for 17,360 result cells. It records one retained algorithmic failure and no integrity failure. Every row identifies the task, source archive, initial design, method, implementation commit, verifier, source/search/verification calls, and postdecision truth reconstruction.

Tables and figures in this manuscript are generated only from compact JSON and CSV analyses. The renderer does not read checkpoints, pickle files, model weights, or exported policy vectors. Runtime artifacts are unnecessary for auditing the displayed results. The method specification, randomized task law, Energy data contract, proof receipt, and rendering manifest are versioned separately so that a theory version cannot be silently paired with another experiment version. The review-V2 geometry, task-strata, and outcome-adjusted-cost supplement is hash-bound to the frozen evidence registry and cannot change a primary outcome.